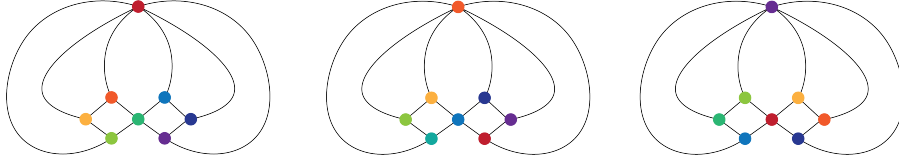


Matroid Intersection: Generalizing Problems in Combinatorial Optimization

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1 Introduction

Matroids are an abstraction of the concept of linear independence from linear algebra. There are many applications of matroids, but one particularly famous application is the matroid intersection. Below are interesting example problems with matroids.

1.1 Colorful Weights

Given a set of elements, each with some weight and unique color, find the maximum weight that can be obtained by selecting a subset of elements such that no element in the subset has the same color.

1.2 Colorful Graph

Given a graph with each edge assigned a color, find a minimum spanning tree where all edges are distinct.

2 Definitions

A **matroid** $M = (S, \mathcal{I})$ is a set S with a collection of subsets $\mathcal{I}$ such that the axioms below hold:

1. For $J \in \mathcal{I}$ and $I \subseteq J$ then $I \in \mathcal{I}$.
2. Given $I, J \in \mathcal{I}$ with $|I| > |J|$, there exists element $x \in I \setminus J$ such that $J \cup \{x\} \in \mathcal{I}$.
3. The empty set $\emptyset$ is in $\mathcal{I}$.

2.1 Terminology

A set I is called **independent** if $I \in \mathcal{I}$ and otherwise **dependent**.

The set S in $M = (S, \mathcal{I})$ is called the ground set of M .

A **basis** is an independent set with maximal size. Notice that every basis has the same size due to Axiom 2, above. If there exists a bigger independent set, one can include the element from it to achieve the same size. Knowing the basis can completely characterize the matroid.

A **circuit** is a dependent set which becomes independent if any element is removed. Note that given two circuits in a matroid that share an element but are not equal, a circuit with elements from both that does not include the shared element can be constructed. Circuits can also characterize a matroid.

An **independence oracle** is a function which determines whether a subset is independent or not in a given matroid. This is used to calculate the time complexity of algorithms involving matroids instead of the regular asymptomatic notation, because the time taken to test independence depends on the matroid. However, it still is useful to characterize how fast an algorithm runs.

A **rank** of a matrix is a function r such that $r(A)$ for some $A \subseteq S$ gives the size of maximum independent subset of A . There are a few properties that will be useful later:

- The rank of a set cannot be greater than its size.
- The rank of an independent set is its size.
- The rank of any subset of S would not be greater than $r(S)$.
- Including an element increases the rank by 1 or keeps it the same. Formally, $r(S) \leq r(S \cup \{x\}) \leq r(S) + 1$.
- Finally, $r(A) + r(B) \geq r(A \cup B) + r(A \cap B)$. Intuitively, this means that it is not always better to put all your eggs in one basket.

2.2 Examples

Let's start with the simplest possible matroid: a uniform matroid. This consists of a ground set S , and any subset $A \subseteq S$ is considered independent if $|A| \leq k$ for some k . The trivial matroid is the case when $k = 0$, that is no set beside the empty set is independent. The complete matroid is when $k = |S|$, that is every subset of S is independent. In this matroid, the basis would be any subset of size k and the circuit would be any subset of size $k + 1$. The rank function would simply be $r(S) = \min(|S|, k)$.

Another simple example is the colorful matroid. It consists of elements, each of which has a color, not necessarily different from the colors of other elements associated with it. A subset is considered independent if it does not contain elements of the same color. We see this follows Axiom 1, because a set with distinct colors will still have distinct colors if you take some subset of them. It follows Axiom 2 because given $|I| > |J|$ there must be some color that we have not used in I . Here, a basis would contain all the distinct colors available in the matroid. The circuit is

more interesting and will consist of two elements of the same color. (Note that it is really easy to see here why a dependent set must contain a circuit.) The rank function would simply count the number of distinct colors in the matroid.

A graphic matroid has a ground set that is a set of edges in some graph, and a subset that is considered independent if it does not contain a cycle. It follows Axiom 1 as it is impossible to introduce a cycle by removing edges. Proving that Axiom 2 is satisfied is harder, but it should be intuitive upon trial and error. The basis of the graphic matroid is set of $n - 1$ edges which connect all n vertices of the graph, called the minimal spanning tree. The circuit would be any cycle in the graph. The rank function becomes very simple as it only returns one less than the number of vertices in the graph (that is, if the graph is connected).

A linear matroid has a ground set of vectors in a vector space, and they are all independent if they are linearly independent. This is where the theory of a matroid originates from, along with the graphic matroid. Any basis of the vector space and the rank of the matroid also corresponds to the rank of the vector space. The circuit does not have any analogue in linear algebra, however.

A truncated matroid can be obtained from any matroid by simply restricting the maximal size of an independent set to be k . For example, we could take a colorful matroid with n colors and truncate it by taking the maximal independent set to be k . Then, any set would be independent if it doesn't contain any duplicate colors and contains at most k distinct colors. Intuitively, this is possible because the subset property of independence does not break when taking the maximum, and the second property of picking a greater independent set becomes infeasible if one didn't exist.

3 Rado-Edmond Algorithm

Consider the problem of finding an independent set with maximal weight. We assign a weight $w : S \rightarrow \mathbb{R}$ to each of the elements in our ground set and $w(I) = \sum_{e \in I} w(e)$. We seek to find $\max_{I \in \mathcal{I}} w(I)$. This is Problem 1.1, Colorful Weights, at the beginning of this article.

3.0.1 Claim

The maximum can be achieved by a greedy algorithm as follows:

1. Sort each of element e in S in descending order of $w(e)$.
2. Start with $I = \emptyset$ and include e in I if $I \cup \{e\}$ is independent.

We see that it takes $\mathcal{O}(n)$ oracle calls and $\mathcal{O}(n \log n)$ operations.

3.0.2 Proof

We will prove this by induction. We see that $\emptyset$ is optimal for size 0. Let A be optimal for independent set size n and B be optimal for $n+1$. Then, by Axiom 2, we have an element $x \in B \setminus A$ such that $A \cup \{x\}$ is independent. Since A and B are both optimal,

$$\begin{aligned} w(A \cup \{x\}) &\leq w(B) \\ w(B \setminus \{x\}) &\leq w(A) \\ w(B) &\leq w(A \cup \{x\}) \\ w(B) &\leq w(A \cup \{x\}) \leq w(B) \end{aligned}$$

Therefore $w(B) = w(A \cup \{x\})$. Thus, for $w(B)$ to be minimal we need $w(x)$ to be minimal.

4 Matroid Intersection

The problem of matroid intersection is given two matroids $M_1 = (S, \mathcal{I}_1)$ and $M_2 = (S, \mathcal{I}_2)$ with the same ground set S , we wish to find a set I with maximum size such that $I \in \mathcal{I}_1 \cap \mathcal{I}_2$. Given this, we see that Problem 1.2, Colorful Graph, reduces to finding the intersection of a colorful matroid and a graphic matroid. Other examples include maximal bipartite matching (where we take each of the matroid to be the sets independent if the set of edge doesn't share a vertex on one side).

4.1 Strong Basis Exchange

To find an algorithm that performs this is quite a challenge. Therefore, we must first make some observations. The **Strong Basis Exchange** states that given two bases A and B , it is possible to pick elements $x \in A \setminus B$ and $y \in B \setminus A$ such that $A \setminus \{x\} \cup \{y\}$ and $B \setminus \{y\} \cup \{x\}$ are both bases. This can be extended to any independent sets $|I| = |J|$ since they can be treated as a basis in the truncated matroid.

Before we prove this, we must build some intuition on how that might work. We see that in a colorful matroid, we can do this by either picking a color that is in $I \setminus J$ and a color that is in $J \setminus I$ or, if it is not possible, then all the colors are the same, so we could swap two elements that have the same color.

To do this we define $\text{span}(A) = \{x \in S \mid r(A) = r(A \cup \{x\})\}$. This is analogous to the definition of span in linear algebra. Now intuitively,

- If $A \subseteq B$ then $\text{span}(A) \subseteq \text{span}(B)$
- If $e \in \text{span}(A)$ then $\text{span}(A) = \text{span}(A \cup \{e\})$

These can obviously be proven rigorously but should be intuitive, although how to prove them might not be obvious.

Using these, we prove the Strong Basis Exchange lemma, which is the following:

4.1.1 Claim

Given two bases A and B , it is possible to pick elements $x \in A \setminus B$ and $y \in B \setminus A$ such that $A \setminus \{x\} \cup \{y\}$ and $B \setminus \{y\} \cup \{x\}$ are both bases.

4.1.2 Proof

Suppose $x \in A \setminus B$. Since B is a basis, $B \cup \{x\}$ must contain a unique circuit, call it C . Now this circuit C must contain x since all the subset of B is independent.

Since C is a circuit, we have $r(C) = r(C \setminus \{x\})$. This further implies that, because

$$C \setminus \{x\} \subseteq (A \cup C) \setminus \{x\}$$

and therefore

$$\text{span}(C \setminus \{x\}) \subseteq \text{span}((A \cup C) \setminus \{x\}),$$

we have $x \in \text{span}(C \setminus \{x\})$ because $r(C) = r(C \setminus \{x\})$. Then, we can conclude that $x \in \text{span}(C \setminus \{x\})$. Now, we have that

$$\text{span}((A \cup C) \setminus \{x\}) = \text{span}(A \cup C) = S,$$

where the last step follows since A is a basis; that is, adding any element does not change its rank.

Then $\text{span}((A \cup C) \setminus \{x\})$ must contain a basis, call it X ; otherwise it cannot span S as it was shown to be. Then $A \setminus \{x\}$ is independent, as any subset of an independent set is, and X is independent with $|X| > |A|$, since all basis have the same size. So, there exists $y \in X \setminus (A \setminus \{x\})$ such that $A \setminus \{x\} \cap \{y\}$ is independent. But we have

$$X \setminus (A \setminus \{x\}) \subseteq \text{span}((A \cup C) \setminus \{x\}) \setminus (A \setminus \{x\}) \subseteq C \setminus \{x\},$$

so in fact y must also be in $C \setminus \{x\}$. Now recall that C was a circuit within $B \cup \{x\}$ so $C \setminus \{x\} \subseteq B$. Since C was the unique circuit in $B \cup \{x\}$, $B \setminus \{y\} \cup \{x\}$ must also be independent because removing y breaks the circuit introduced by adding $\{x\}$.

4.2 Main Algorithm

For a matroid $M = (S, \mathcal{I})$ and $I \in \mathcal{I}$, the exchange graph $D_M(I)$ is defined as a **bipartite graph** between I and $S \setminus I$ with edges between $x \in I$ and $y \in S \setminus I$ if $I \setminus \{x\} \cup \{y\}$ is independent.

Now if I and J are both independent with $|I| = |J|$, then there exists a matching between $I \setminus J$ and $J \setminus I$ in $D_M(I)$. By the Strong Basis Exchange, we can find an edge in $D_M(I)$ between $I \setminus J$ and $J \setminus I$ and updated I to $I' = I \setminus \{x\} \cup \{y\}$ and $J' = J \setminus \{y\} \cup \{x\}$. This can continue until the sets are the same.

However, the converse is not true, and it is easy to come up with a counter example. To fix this, we need to add a condition specifying that matching must be unique. This happens to be a necessary and sufficient condition for an independent set.

We can also combine the exchange graph for both M_1 and M_2 . The directed bipartite graph $D_{M_1, M_2}(I)$ for $I \in \mathcal{I}_1 \cap \mathcal{I}_2$ has bipartitions I and $S \setminus I$ and has edge from I to $S \setminus I$ when $I \setminus \{y\} \cup \{x\} \in \mathcal{I}_1$ and edge from $S \setminus I$ to I when $I \setminus \{x\} \cup \{y\} \in \mathcal{I}_2$.

Now we can finally state the algorithm to find the intersection:

1. Set $I = \emptyset$.
2. Find $X_1 = \{x \notin I \mid I \cup \{x\} \in \mathcal{I}_1\}$ and $X_2 = \{x \notin I \mid I \cup \{x\} \in \mathcal{I}_2\}$.
3. Find an augmenting path in $D_{M_1, M_2}(I)$, a path from an element in X_1 to an element in X_2 which does not contain shortcuts (edge that point from an earlier vertex on the path to a non-adjacent later vertex on the path).
4. If there is no such path, I is the maximal intersection.
5. Otherwise, we replace I with $(I \setminus P) \cup (P \setminus I)$ and go back to Step 2.

4.2.1 Claim

The output I is independent in both $\mathcal{I}_1$ and $\mathcal{I}_2$.

4.2.2 Proof

We can assume that at the start I is an independent set in both M_1 and M_2 . Then, we want to prove $(I \setminus P) \cup (P \setminus I)$ is also an independent set in both M_1 and M_2 .

First, we show that $I \in \mathcal{I}_1$. We define $M'_1 = (S \cup t, \{J \mid J \setminus \{t\} \in \mathcal{I}_1\})$ or in standard terms, M_1 with added element t which is independent to all the elements in M_1 . We can define M'_2 analogously. Then, we know the path P combined with t creates a matching between $I \cap P \cup \{t\}$ and $S \setminus I$. This matching is unique since P does not contain any shortcut. Therefore, $(I \setminus P) \cup (P \setminus I)$ is independent in $\mathcal{I}_1$.

Similarly, I is independent in $\mathcal{I}_2$.

4.2.3 Claim

The output I is largest I possible.

4.2.4 Proof

To prove this, we will show that

$$\max_{I \in \mathcal{I}_1 \cup \mathcal{I}_2} |I| = \min_{U \subseteq S} r_1(U) + r_2(S \setminus U).$$

We see that for any $U \subseteq S$ and $I \in \mathcal{I}_1 \cup \mathcal{I}_2$,

$$\begin{aligned} |I| &= |I \cap U| + |I \cap (S \setminus U)| \\ &\leq r_1(U) + r_2(S \setminus U) \end{aligned}$$

since $I \cap U \in \mathcal{I}_1$ and $I \cap (S \setminus U) \in \mathcal{I}_2$.

If we could come up with $U \subseteq S$ and $I \in \mathcal{I}_1 \cup \mathcal{I}_2$ such that the equality was satisfied, then the maximal size of I would be $|I|$ since anything greater will violate the condition that $|I|$ should be less than the minimum with a given U .

If we set $U = \{z \in S \mid z \text{ has a path to a vertex in } X_2\}$, then we claim that $r_1(U) = |I \cap U|$ and $r_2(S \setminus U) = |I \cap (S \setminus U)|$.

If $r_1(U) \neq |I \cap U|$, then since $I \cap U \subseteq U$ and $I \cap U \in \mathcal{I}_1$, we can see that $r_1(U) > |I \cap U|$. Then by the Axiom 2, there exists $x \in U \setminus I$ such that $(I \cap U) \cup \{x\} \in \mathcal{I}_1$. Then we have that either $I \cap \{x\}$ is independent or contains a unique cycle C which contains at least one element not from $(I \cap U) \cup \{x\}$. Either way, we have $y \in I \setminus U$ such that $I \setminus \{y\} \cap \{x\} \in \mathcal{I}_1$ so there exists a directed edge from y to x . Then $y \in U$, since $x \in U$, which is a contradiction.

We can similarly prove that $r_2(S \setminus U) = |I \cap (S \setminus U)|$.

5 Conclusion

There are countless applications for matroids. Matroids are fundamental in the study of combinatorial optimization, and some other practical techniques include matroid unions and matroid partitions. Furthermore, there are other areas of research in matroids, such as whether a matroid can be represented as a matrix.

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